

Model C 315/350

CIRCULAR CUTOFF SAW

OPERATORS MANUAL

PARTS LIST

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Specifications

Model	**C 315/350**
Drive Motor	**3.5 HP, 2 speed, TEFC**
Spindle RPM	**30 & 60**
Blade Data	
Max diam	**14" (350mm)**
Spindle diam	**32mm**
Pin diam	**10mm (2)**
Pin locations	**63mm**
Max Vise Opening	**6.25"**
Work Height	**39 1/2"**
Power Requirements	
	208-220-230 volts - 20 amps
	440-460 volts - 12 amps
	3 phase
	60 cycles
	Note: two speed motor is not dual voltage
Coolant Capacity	**1.75 Gal (8 lit)**
Gearbox Oil Capacity	**14oz (.5 lit) 90 weight gearlube**

SAFETY INSTRUCTIONS

TO THE OWNER OR SUPERVISOR

You must remember that this machine is designed to cut metal with a sharp tool, and you are responsible to see that the machine is in top operating condition and that it is operated IN A SAFE MANNER!

YOU MUST:

1. Make sure the machine is properly installed, anchored to floor, and electrical installation is proper.
2. Be sure you are familiar with all operating, safety, and applications information before operating this machine or turning it over to one of your employees.
3. See that all who operate this machine are properly trained and fully aware of all safety practices.
4. Be aware of all unsafe practices that may occur. (See operator safety precautions and applications information)
5. Insist on proper personal protective equipment and practices.
6. Maintain all factory installed safety devices and insure that these are never removed or altered or restricted in any way.
7. Insure that your operators have a safe and orderly work area, with adequate light and operating room.
8. Be certain that your machinery receives responsible and competent maintenance and that your machinery is inspected on a regular basis.

YOU ARE RESPONSIBLE

SAFETY INSTRUCTIONS

TO THE MACHINE OPERATOR

You must read and understand all operating, safety, and applications information before attempting to use this machine.

1. The machine is provided with double vises to enable you to clamp the work on both sides of the blade. This is important in obtaining the best quality cut, and it is **CRITICAL** for **SAFE** operation.
2. Always wear proper eye protection.
3. Never operate machine unless all guards are in place.
4. Never insert hands or arms into or near cutting area while machine is running.
5. Never load or unload machine while blade is running.
6. Never wear loose clothing, long sleeves, gloves, jewelry, or any other items that may be caught. Confine long hair.
7. Adequately support stock on both sides of machine to prevent falling.
8. Always disconnect power at source when performing maintenance or making adjustments, other than those necessary for the normal operation of machine.
9. Keep cutting area clear of tools or other loose objects.
10. Before starting a sawing cycle, be sure BOTH vises are securely clamped and machine set-up is correct.
11. Never use this machine to cut workpieces larger than the stated capacity.
12. Never modify a blade to fit arbor.
13. Never use a blade larger than the machine is designed for.
14. When cutting very short pieces, make sure they do not jam between the blade and the slot in the tabletop.
15. Be aware that the pneumatic vises close automatically and to KEEP HANDS CLEAR! (Model C315 AV only)
16. Be aware of the possible sawing problems (see APPLICATIONS section of this book)
17. CAUTION: If blade becomes jammed, immediately turn off power, loosen vise, extract workpiece from blade, and inspect blade for cracks or broken teeth.

ALWAYS OPERATE MACHINE SAFELY, USING COMMON SENSE AND ALERTNESS

INSTALLATION

We recommend that you follow these instructions closely and report any problems to your distributor.

Select an installation site that is clear of traffic and obstructions, suitable for proper machine operation and maintenance, including adequate light and clearance.

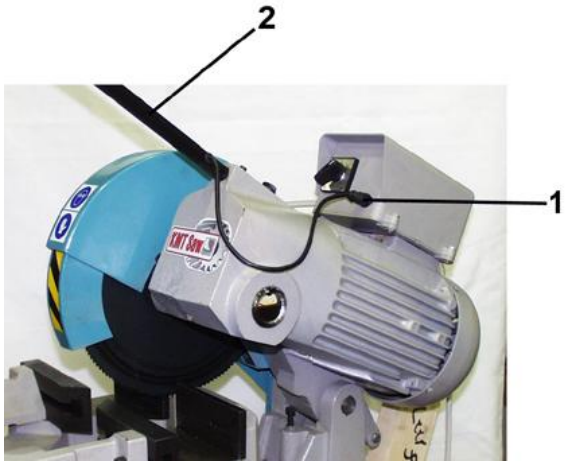
The first step after uncrating your machine is to see that all items are present and that there is no apparent damage to any part of the machine.

Checklist of items

C 315/350 Machine and Base
Operators manual
Wrenches
1 saw blade, 275mm or 300mm (unless you ordered something different than standard)
Stock Stop Assembly
Operating & Switch Handle

Your machine has been carefully packed and has preservative applied to all metal surfaces to protect it during transit and storage. It is recommended that all of these surfaces be cleaned at this time.

Your machine gearbox has been properly filled prior to shipment. The site glass is merely to allow you to determine at some later time if oil has leaked. At minimum oil should be visible halfway in site window.



Remove saw frame bracket and install the operation & switch handle assembly (#2) to the saw frame, and secure. Plug into electrical box (#1).

Mix coolant to approximately 10:1 with water for most sawing applications. For Stainless Steal use a 5:1 mixture. For unusual materials, consult the manufacturer for additional coolant recommendations.

Mix coolant into water in order to mix more easily. Pour coolant directly into the machine base.
Never use straight cutting oil!

ELECTRICAL CONNECTION

Only a qualified electrician should do electrical connections!

Make sure your power supply matches the machine. All machines are designed for a specific operating voltage and cannot be changed.

It is recommended that the machine is on a circuit that is **properly fused,** and the machine must be **properly grounded** to comply with OSHA and state electrical codes.

Connect machine cord to power supply. Note that the green/yellow wire is ground.
After connecting to 3-phase supply, check the machine for proper rotation. If rotation is not correct, change any two of the three power leads at point of connection.

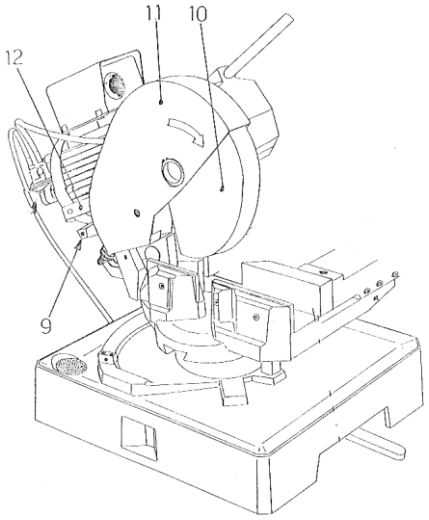
WIRE SIZES

The machine must be serviced by adequate wire or a severe loss of power may occur, leading to poor machine performance. We recommend:

Length of conductor	Wire Gauge Size
50' or less	#14
100" or less	#12
100-150"	#10
150 or longer	NOT RECOMMENDED

Wiring Diagram Enclosed

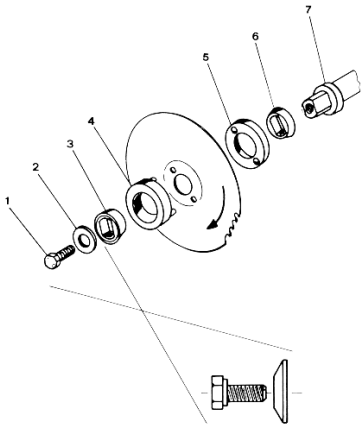
BLADE INSTALLATION



Please refer to the section on **APPLICATIONS** to select proper blade.

Remove screw (9) allowing guard (10) to swivel back inside guard (11) while lifting operating lever. (12)

-Remove spindle bolt (LH Thread) and remove the clutch/flange assembly (items 1 thru 6)
-Insure that the spindle, outer flange, and blade are clean and free of dirt and oil. When clean, lubricate with only a drop of oil.



- Remount items 5 & 6.

-Mount the blade with the teeth pointing in the direction of the arrow.

-Mount part 4, aligning pins and holes.

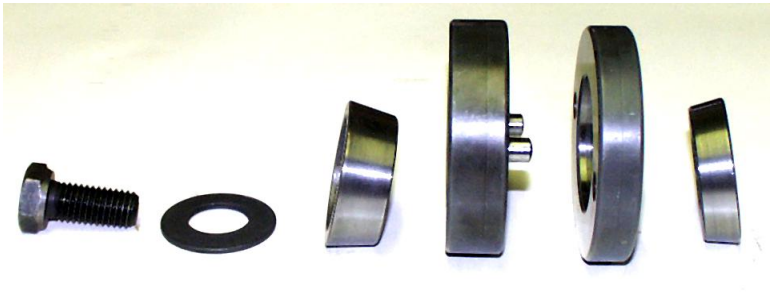
-Mount part 3, then the spring washer 2 (**high center side to the outside**), and the spindle bolt (1).

IMPORTANT: Prior to finally tightening the spindle bolt, rotate the blade backwards so the backlash is removed from between the driving pins and the holes in the blade.

IMPORTANT: Prior to finally tightening the spindle bolt, rotate the blade backwards so the backlash is removed from between the driving pins and the holes in the blade. Tighten spindle bolt snugly, then turn additional 3/4 turns. If during the course of sawing the blade stalls while the spindle continues to turn, apply 1/4 turn more on the spindle bolt (see **CLUTCH** below)

-Reinstall the blade guard.

CLUTCH
The clutch of your machine is extremely important in terms of blade life, blade protection, and transmission of power to your workpiece.
Prior to making a single cut and before putting on a blade, you should remove the clutch assembly, clean thoroughly, and apply a light coating of oil.



If the blade stops easily in the material during the course of sawing, it may be necessary to adjust the clutch by retightening spindle bolt, or your clutch may need to be removed and cleaned. It is also very important to make sure that the problem is not a misapplication or dull blade causing the problem. A dull blade or a blade not properly suited for the application may also easily stall in the material.

OPERATING THE MACHINE

Install stock stop assembly if use desired.



-Adjust depth stop (1) so the blade will cut completely through the workpiece.
-Adjust the four adjustable vise jaw inserts as closely as possible to the blade for maximum support to the workpiece.
-Insure locking lever (2) is swiveled fully to the right.
-Turn vise handwheel to clamp vise on workpiece. Start blade motor by turning main switch to "ON" position squeeze handle switch, and adjust coolant flow with valve on top of blade guard.
-When ready to make a cut, set selector switch for blade speeds to the desired setting and complete a cut by grasping the operating lever and bringing the saw through the workpiece.

ANGLE CUTTING

Your machine will make angle cuts from 0-45° left or right. It will be necessary to readjust the vise jaw inserts on the moving vise jaws when miter cutting, as failure to do so will result in interference with the blade. First loosen locking lever (#2), and rotate saw frame in the desired direction until the pointer coincides with the angle desired. Relock lever.



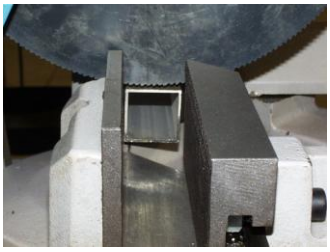


VISE DOVETAIL

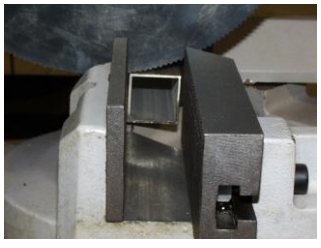
It is important to notice that the entire vise assembly will move laterally on a dovetail. (closer to the sawframe pivot or closer to the saw operator) This is adjusted by unlocking lever (#2) and sliding the assembly to the most appropriate position; closer to the pivot, or closer to operator.

The most advantageous position is normally so that the sawblade will going through the smallest possible cross section at any time. This may not always be the position that allows you to get through larger workpieces. Examples below show the INORRECT and CORRECT positions. If sawing small solid materials, place the center of the workpiece slightly behind spindle center.

Also note that changing this position might change the cutoff point of the sawblade (#1 above)



INCORRECT



CORRECT

Blade Selection

It is recommended in all cases to **use the smallest sawblade possible at all times.**
This will enhance the machines ability to cut, it will make it easier for the machine operator to cut, and it will increase the life of your machine. In addition, you may consider the blade to be similar to a pulley; the smaller the diameter, the greater the torque. Normally smaller blades will cut with more power, less vibration, and you can cut with greater ease.

The 14" sawblade is intended only for occasional maximum capacity purposes, and never intended to be used for production. Extended use of 14" sawblades will decrease gearbox life.

Blade Spindle Speeds

Your machine has two spindle speeds. You may select the correct spindle speed as follows:

Fast - #2 on selector switch (60 rpm)
To be used **only for thin structurals & aluminum** or other non-ferrous materials

Slow - #1 on selector switch
All other jobs

Blade Toothpitch

Refer to following chart to locate the type of material (solid or structural) and size of material you want to cut.

Locate material to be cut, and follow straight down to the row labeled "**toothpitch**" to find recommended toothpitch.

MATERIAL AND PITCH SELECTION CHART					
STRUCTURALS □		○ ⌴			
mm	2	3 - 6	6 - 10	10 - 12	12 - 14
in	0.080	.125 - .250	.250 - .400	.400 - .500	.500 - .550
ga	13	11 - 8	-	-	-
fraction	5/64	1/8 - 1/4	1/4 - 3/8	3/8 - 1/2	1/2 - 9/16
SOLIDS					
mm	10	20	30	40 - 60	60 - 80
in	0.400	0.750	1.250	1.5 - 2.5	2.5 & larger
Model C315/350 ■		●			
toothpitch	3	4 - 5	6	8	not
					recommended

Locate recommended blade from list of blades on following chart.

CIRCULAR SAW BLADES LIST								
HIGH SPEED STEEL BLADES, OXIDE COATED, RC 63/65								
Arbor: 32mm - Pinholes: 2 x 11 x 63mm								
Blade #	Size	Pitch	# of Teeth		Blade #	Size	Pitch	# of Teeth
1002	250mm (10")	2	390		1303	315mm (12 1/2")	3	310
1003	250mm (10")	3	240		1304	315mm (12 1/2")	4	250
1004	250mm (10")	4	180		1305	315mm (12 1/2")	5	190
1005	250mm (10")	5	160		1306	315mm (12 1/2")	6	160
1006	250mm (10")	6	140		1307	315mm (12 1/2")	7	150
1007	250mm (10")	7	110		1308	315mm (12 1/2")	8	130
1103	275mm (11")	3	280		1310	315mm (12 1/2")	10	110
1104	275mm (11")	4	220		1312	315mm (12 1/2")	12	90
1105	275mm (11")	5	180		1403	350mm (14")	3	360
1106	275mm (11")	6	140		1404	350mm (14")	4	280
1107	275mm (11")	7	120		1405	350mm (14")	5	220
1108	275mm (11")	8	110		1406	350mm (14")	6	180
1110	275mm (11")	10	90		1407	350mm (14")	7	160
1112	275mm (11")	12	70		1408	350mm (14")	8	140
1203	300mm (12")	3	300		1410	350mm (14")	10	110
1204	300mm (12")	4	240		1412	350mm (14")	12	90
1205	300mm (12")	5	180					
1206	300mm (12")	6	150					
1207	300mm (12")	7	140					
1208	300mm (12")	8	120					
1210	300mm (12")	10	100					
1212	300mm (12")	12	80					

Special Sawblade Grinds

All standard blades are ground for use on mild steels.

Blades to be used on stainless steels (or other hard materials) should be designated with an “SS” after the part number (ex: 1204 SS) These blades may also be used on mild steels.

Blades to be used on aluminum or other non-ferrous materials should be designated with an “AL after the part number (ex: 1210 AL) ***These blades may not be used on steels.***

Coolant Pump

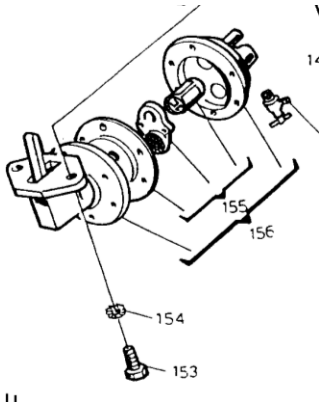
The coolant pump is a diaphragm pump that works automatically when the motor is running. The system contains two filters, that can be checked and washed clean of trash. The first filter is in the tank at the end of the coolant line, and the other is in the pump.

Please note that repair kits are no longer available for these pumps.

Too frequently the repair kit did not solve pump problems.

The complete pump is part number 691300, shown at right.

It does not include on/off petcock.



Operators handle & switch assembly

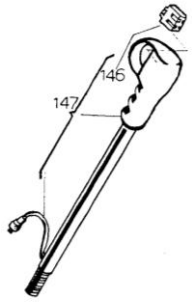
Common detailed parts are:

The complete assembly is part number 132595

The plastic grip is part number 513573

The switch is part number 520821

The plug & cord is part number 204704



Instructions for Gear Replacement
Model C 315/350

1. Drain oil from Gearbox through plug in front (1)

2. Remove 4 screws #7. Loosen screws #8 and remove casting #9.

3. Remove complete assembly #10, and screw #11.

4. Remove guard support #13 & #14.

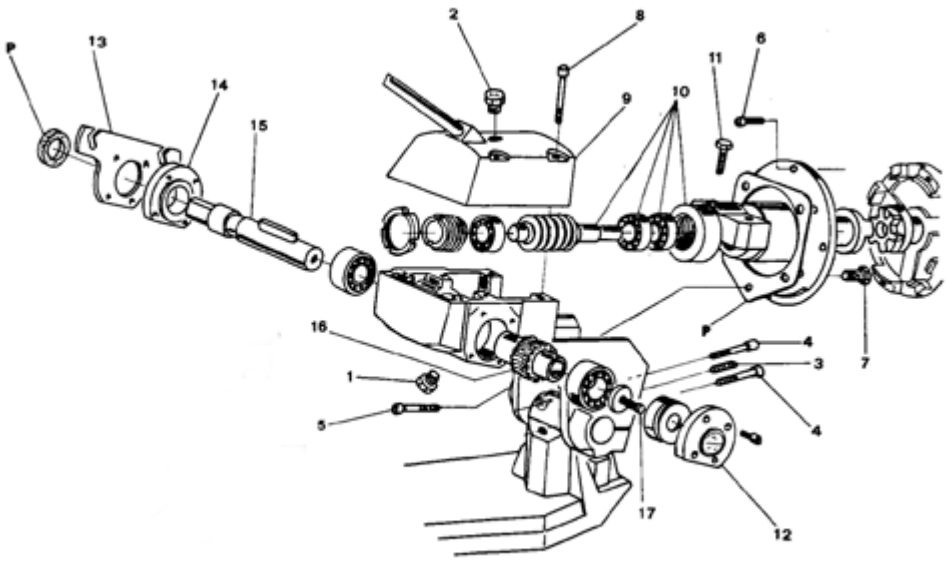
5. Remove site glass #12. Loosen screw #17.

6. Remove shaft #15 by driving it out through the hole that holds site glass.

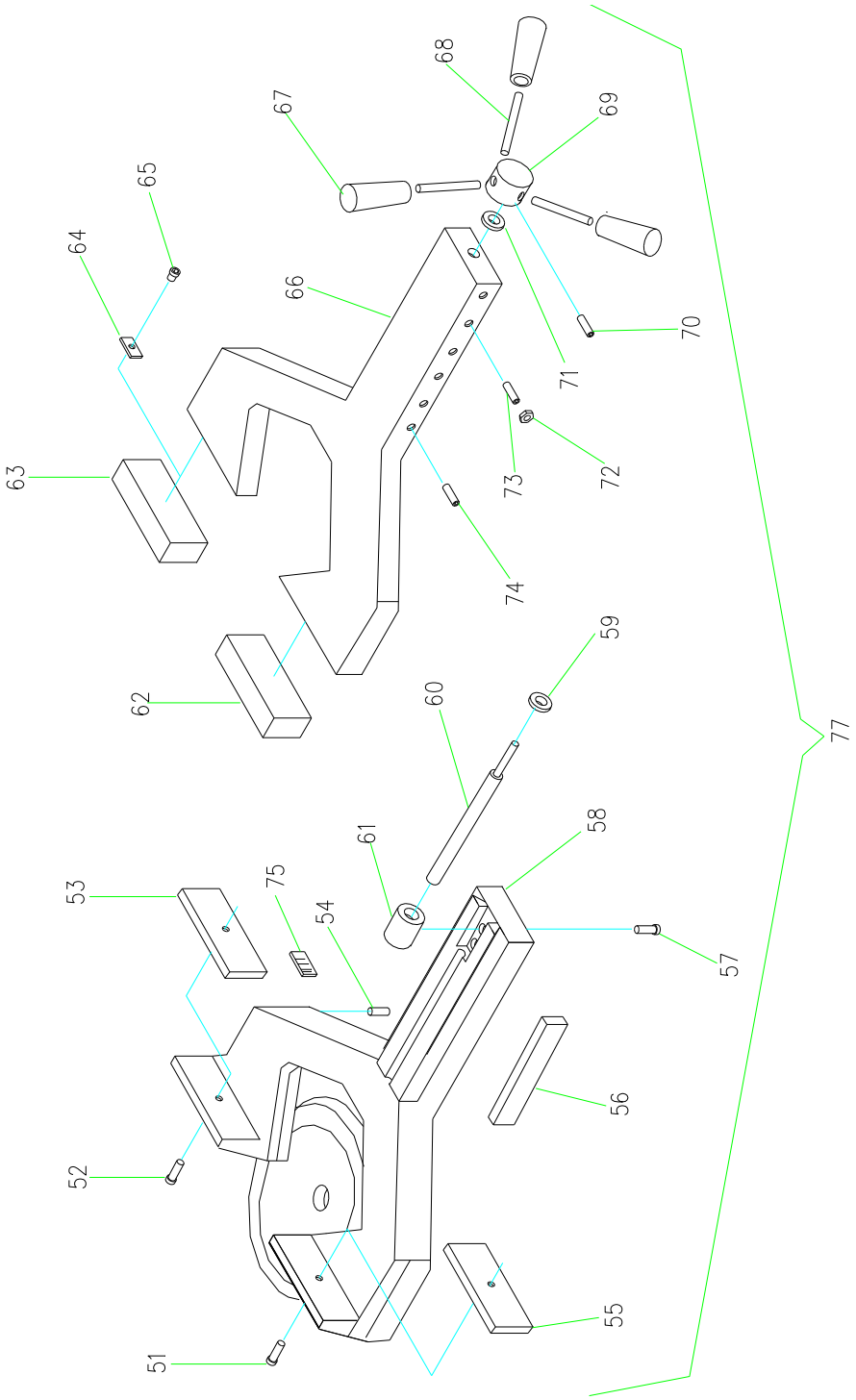
7. Replace bronze gear assembly.

8. Reassemble.

9. When saw has been completely reassembled and remounted to base, add oil with saw head fully up, until oil level is visible in sight glass.



Double Vise Assembly
Replaces single vise assembly on new machines
Typical wear part is item # 61, Part #693005 bronze nut
Caution: Do not overtighten. It's not a bench vise.



Nø	CODICE	DESCRIZIONE	Q	Nø	CODICE	DESCRIZIONE	Q	Nø	CODICE	DESCRIZIONE	Q	Nø	CODICE	DESCRIZIONE	Q
1	558498	MORSA COMPLETA PERFECT 300	1	51	978095	VITE TCE 8X20 UNI5931	1								
2	978870	VITE TCE M8X25 UNI5931	1	52	979645	VITE TCE 8X30 UNI5931	1								
3	183227	CARRELLO MORSA	1	53	471350	GANASCIA APP.DX/SX MORSA	1								
4	864825	TAPPO OLIO 1/8"+GUARN.	1	54	942445	VCE GRANO 12X20B UNI5925	1								
5	182962	CAPPUCCIO PRES. CAP-1 31060	1	55	471713	GANASCIA APP.DX/SX MORSA	1								
6	699897	PRESOSTATO PM0.15-3A T=1 1/8G	1	56	279921	CUNEO REGISTRO MORSA	1								
7	978870	VITE TCE M8X25 UNI 5931	1	57	983893	VITE TCE 10X25 UNI5931	1								
8	749470	ROSETTA PIANA DIAM.8	2	58											
9	775918	SILENZIATORE PN.SCARICO 1/8 NYLON	1	59	741509	RONDELLA APPOGGIO SS20X28X2	1								
10	735520	RONDELLA ALL.1-8	1	60	957104	VITE CHIUSURA MORSA	1								
11	695995	PORTA GOMMA PG6-8	1	61	693005	PORTA VITE TPN25 MORSA DOPPIA	1								
12	716465	RACCORDO GOMITO 5020 1/8"	1	62	478001	GANASCIA CHIUS.MORSA DOPPIA	1								
13	256570	COM.PN. 3/2NA=3LL-8=ALR8	1	63	478002	GANASCIA CHIUS.MORSA DOPPIA	1								
14	747052	ROSETTA D4	2	64	677103	PIASTRINA X GANASCIA 22X22	2								
15	975923	VITE TCE 4X25 UNI5931	1	65	984002	VITE TCE 10X35 UNI5931	1								
16	723120	RACCORDO "T" FFM 4050 1/8"	1	66	184975	CARRELLO MORSA DOPPIA	1								
17	716465	RACCORDO GOMITO 5020 1/8"	1	67	513870	IMPUGNATURA NERA 12X70	3								
18	720020	RACCORDO ARIA GOM.RAP.D8 1/8"	1	68	537422	MANIGLIA D12 MORSE	3								
19	727514	RIDUZIONE 1/8"M-1/8"F	1	69	279921	GHIERA X VITE MORSA	1								
20	720795	RACCORDO ARIA DIR.RAP.D8 1/4"	1	70	790961	SPINA EL.8X50	1								
21	616170	NIPPLO N4-4 1/4" A 1/4" OTT.	1	71	741509	RONDELLA APPOGGIO SS20X28X2	3								
22	968795	VITE VSP M8X25 UNI5923	2	72	306500	DADO ES.M6 UNI5588	1								
23	716145	RACCORDO GOMITO 5020 A 1/4"	1	73	943855	VCE GRANO 6X30C UNI5927	1								
24	775943	SILENZIATORE PN.SCARICO 1/4"	1	74	943903	VCE GRANO 6X35C UNI5927	1								
25	766643	SCARICO RAPIDO ARIA SR/4	1	75	910206	TARGHETTA GRADUATA MORSE	1								
26	935470	TUBETTO ARIA RILSAN 6X8 PA11	1	76	158378	BLOCCO MORSA DOPPIA	1								
27	132693	ASTA COM.24V+PRES. 2BAR	1	77	558358	MORSA COMPLETA DOPPIA	1								
28	494805	GHIERA ESTERNA CILINDRO PN. D210	1												
29	686602	PIATTELLO X CILINDRO PN. D210	1												
30	494407	GHIERA INTERNA CILINDRO PN. D210	1												
31	288097	CUSCINETTO ASS.51104 20X35X10	1												
32	976100	VITE TCE M5X30 UNI5931	1												
33	507621	GUARNIZIONE CIL.MORSA PN. D210	1												
34	344684	FLANGIA CILINDRO PN. D210	1												
35	658290	PIASTRA ATTACCO MORSE PN. D210	1												
36	628570	PALETTA COMANDO MORSA PN.	1												

Tavola NG 3/3

Denominazione ELENCO RICAMBI
MOR.PN.MOR.DOP.RECORD 315/350

Gruppo RECORD 315/350

Data /10/98

Diseg. RCD

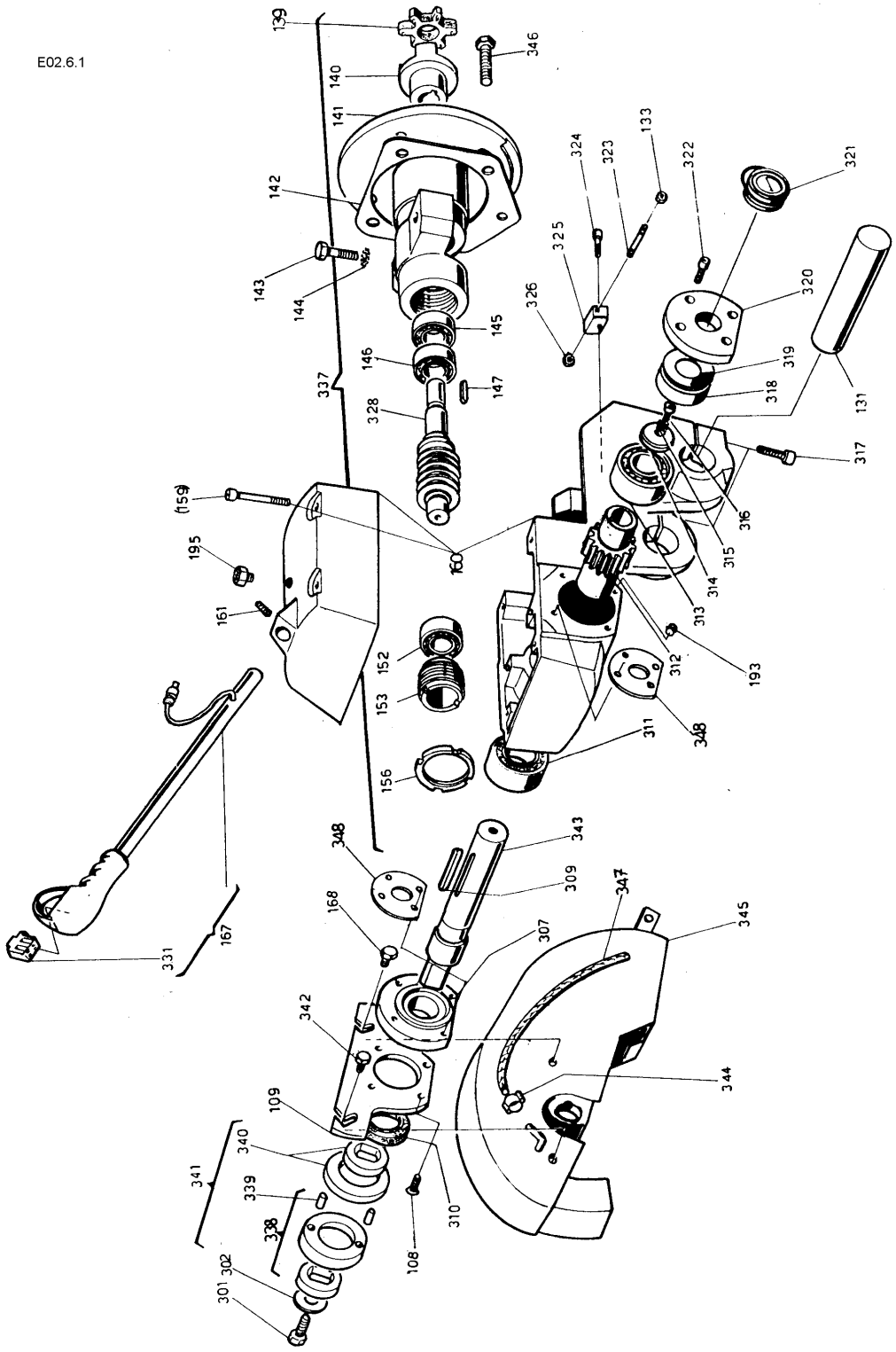
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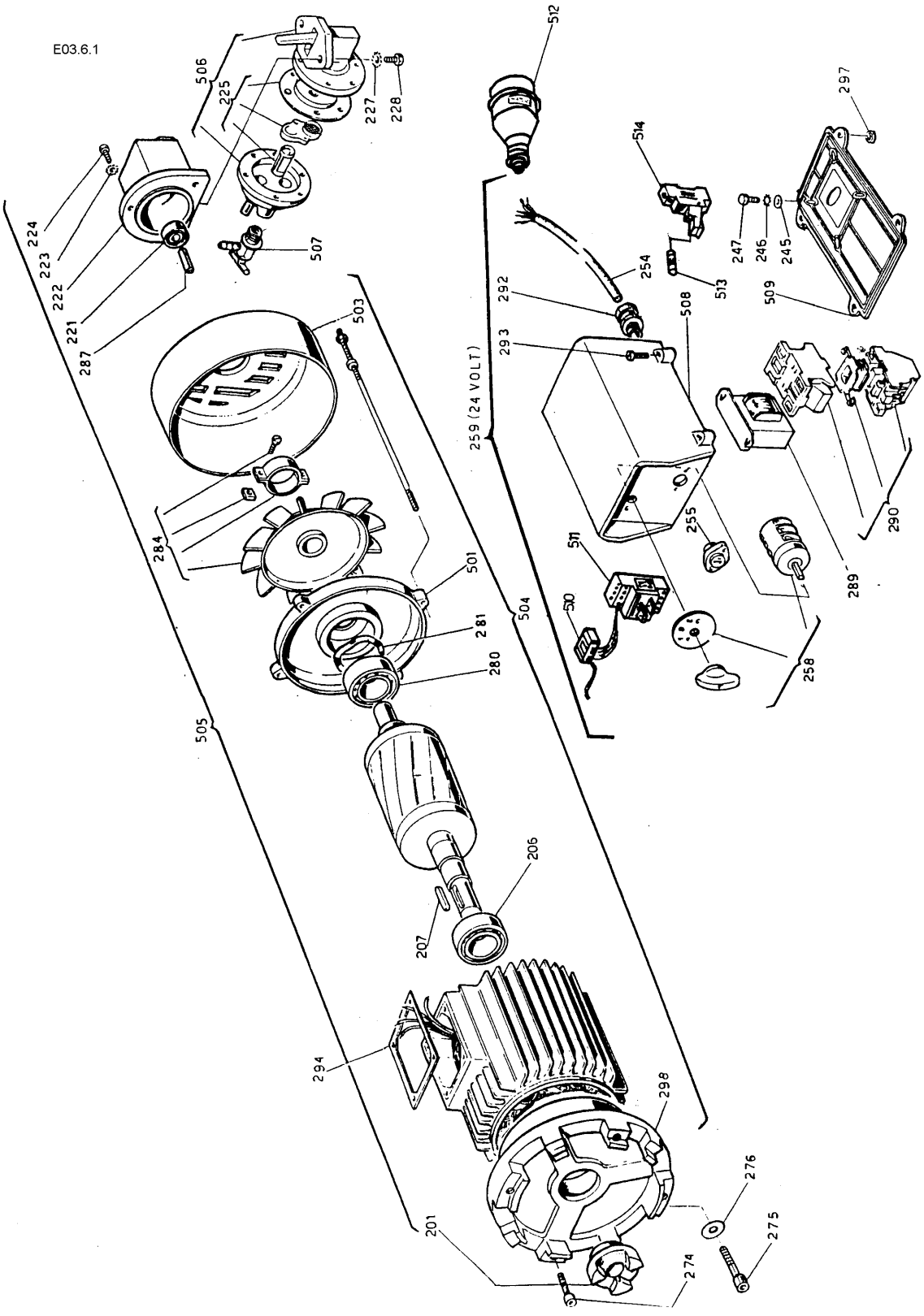
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Nø	CODICE	DESCRIZIONE	Q	Nø	CODICE	DESCRIZIONE	Q	Nø	CODICE	DESCRIZIONE	Q	Nø	CODICE	DESCRIZIONE	Q	
1	528305	LEVA CON ECCENTR. RECORD	1	51	957077	VITE CARRELLO MORSA	1	153	120170	ANELLO FILETTATO MANICOT.VSF	1	314	742431	RONDELLA 35X10X6	1	0
2	750245	ROSETTA EL.D12 DIN6798	1	52	741509	RONDELLA APPOGGIO SS20X28X2	1	156	497595	GHIERA KM11 M55X2	1	315	750600	ROSETTA EL.D10 UNI6798	1	
3	974220	VITE TCE M12X45 UNI5931	1	53	545458	MOLLA RITORNO MORSE D30 FILO4	1	159	983520	VITE TCE M8X70 UNI5931	2	316	984300	VITE TE M10X35 UNI5739	1	
4	941670	VCE GRANO M12X16A UNI5923	1	54	741509	RONDELLA APPOGGIO SS20X28X2	1	160	E92772	TESTA+COPERCHIO RECORD	1	317	981195	VITE TCE M8X40 UNI5931	1	
5	632425	PASTIGLIA D10 NYLON	1	55	499145	GHIERA GN NORM. M15X1 AUTOF.	1	161	942445	VCE GRANO M12X20B UNI5925	1	318	323409	DISTANZIALE CUSCINETTO D62X21	1	
6	301520	DADO M24 REGISTRAZIONE BASAM.	1	56	963837	VOLANTINO MORSE	1	167	132595	ASTA COM.24V COMP.D20	1	319	546622	MOLLA TAZZA 61.5X40.5X0.7	1	1
7	143801	BASAMENTO RECORD	1	57	288097	CUSCINETTO ASS.51104 20X35X10	1	168	961500	VITE CARTER M12X15 POST.	1	320	349678	FLANGIA POSTERIORE MANDRINO D45	1	
8	335748	FILTRO PESCANTE NYLON D12 K58	1	58	694083	PORTA CUSCINETTO MORSA	1	193	865936	TAPPO OLIO 1/4"=GRANO DIN906	1	321	786162	SPIA OLIO HE45	1	
9	936245	TUBETTO GEMMA 8X12	1	59	529303	LEVA ECCENTRICA MORSA	1	195	865720	TAPPO+GURNIZ.1/4 TKTG13 NOBFL	1	322	978095	VITE TCE M8X45 UNI5931	4	
10	505580	GRIGLIA CON TUBETTO BASAMENTO	1	60	183483	CARRELLO MORSA PERFECT 300	1					323	646500	PERNO ATTACCO MOLLE L85 M8	1	
11	994370	VITE TE M6X10 UNI5739	2	61	306500	DADO ES. M6 UNI5588	8	201	774270	SEMIGIUNTO ELAST. OTTONE	1	324	981970	VITE TCE M8X45 UNI5931	1	
12	505070	GRIGLIA 72X72 BASAMENTO	1	62	943807	VCE GRANO M6X25C UNI5927	8	206	290670	CUSCINETTO 62062RS 30X62X16	1	325	145072	BLOCCHETTO ATTACCO MOLLE	1	2
13	354220	FORCELLA ATTACCO MOLLE	1	63	928673	TIRANTE M10 GANASCIA SCORREVOLE	1	207	214720	CHIAVETTA 6X6X25 UNI6604	1	326	307720	DADO ES.M8 UNI5588	1	
14	179636	BUSSOLA DISTANZIALE D8X12X20	3	64	677103	PIASTRINA X GANASCIA 22X22 M10	2	221	286020	CUSCINETTO POMPA/SPAZZ.608ZZ	1	328	960515	VITE SENZA FINE	1	
15	983550	VITE TCE M8X90 UNI5931	1	65	970273	VITE TCE BOMBATA M8X12	1	222	771945	SCATOLA PORTA POMPA	1	331	520821	INTERRUTTORE ASTA 24V AP65	1	
16	294189	DADO ES.M8 ALTO AUTOBL.DIN980	1	66	169002	BRACCIO ANTIBAVA	1	223	751020	ROSETTA EL.D5 DIN6798	1	337	533375	MANICOTTO+VSF M2.75	2	
17	750011	ROSETTA SCHNORR D8 ZIG.	3	67	974995	VITE TCE M12X50 UNI5931	1	224	992045	VITE TE 5X20 UNI5739	1	338	341897	FLANGIA+ANELLO A CONO CHIUSURA	1	
18	978095	VITE TCE M8X20 UNI5931	1	68	738976	RONDELLA 45X12X5	1	225	727050	RICAMBIO POMPA	1	339	791880	SPINA CILINDRICA RETT.7.97X18	2	3
19	549106	MOLLA RITORNO TESTA FILO D4	1	69	984320	VITE TE M10X40 UNI5739	1	227	751795	ROSETTA EL.D8 DIN6798	2	340	341726	FLANGIA+ANELLO A CONO APPOGGIO	1	
20	983520	VITE TCE M8X70 UNI5931	2	70	478286	GANASCIA X ANTIBAVA	1	228	995145	VITE TE 8X20 UNI5739	2	341	506862	GRUPPO FRIZIONE A CONO COMPL.	1	
21	300745	DADO ES.M8 RIBASSATO	1	71	961056	VITE ECCENTRICA PER MORSA	1	245	744611	RONDELLA STAMPATA 5X15X1.2	1	342	961500	VITE CARTER M12X15 POSTERIORE	2	
22	946320	VCE GRANO M8X35B UNI5925	1	72	472350	GANASCIA APPOGGIO SINISTRA	1	246	751020	ROSETTA EL.D5 DIN6798	1	343	109999	ALBERO FRESA RECORD	1	
23	138770	ASTINA INTERNA D6.5 PIATT.	1	73	470350	GANASCIA APPOGGIO DESTRA	1	247	970075	VITE TCROCE 4.2X19 AUTOF.	1	344	331888	FASCETTA STRING.STRK11-13	2	4
24	549106	MOLLA RITORNO TESTA FILO D4	1	74	978870	VITE TCE M8X25 UNI5931	1	254	205065	CAVO 4POLI SEZION.1.5 CEI	1	345	191623	CARTER GLOBALE RECORD	1	
25	987395	VITE TE M12X45 UNI5739	1	75	978870	VITE TCE M8X25 UNI5931	1	255	698504	PRESA BIPOLARE X COM.24V	1	346	987105	VITE TE M12X30 UNI5931	4	
26	303845	DADO ES.M12 UNI5588	1	76	978870	VITE TCE M8X25 UNI5931	1	258	257909	COMMUT.12A 1-0-2 TECNOMATIC	1	347	936245	TUBETTO GEMMA 8X12	1	
27	303845	DADO ES.M12 UNI5588	1	77	558496	MORSA RECORD	1	259	253518	COM.EL.24V + SALVAM. V400	1	348	511480	GUARNIZIONE ALBERO FRESA	2	
28	988375	VITE TE M12X80 UNI5739	1	78	134120	ASTA DISCESA TESTA/REG.MORSA	1	274	978870	VITE TCE 8X25 UNI5931	1					
29	978870	VITE TCE M8X25 UNI5931	2	79	723776	REGISTRO MISURA D20 PLASTICA	1	275	981970	VITE TCE M8X45 UNI5931	1	501	263799	COPERCHIO POSTER. MOT.FC100 D172	1	
30	169088	BRACCIO ATTACCO CARTER	1	80	976547	VITE TCE M6X25 UNI5931	1	276	744820	RONDELLA STAMPATA 8X24X2	1	503	271472	COPRIVENTOLA RECORD	1	
31	958402	VITE RIBASSATA CARTER TCE M8X14	1	81	791760	SPINA CIL.RETT.8X24 UNI6364A	1	280	289120	CUSCINETTO 6206ZZ 30X62X16	1	504	585725	MOT. 3F 4-8P FC100+TP V400	1	5
32	630195	PARASPRUZZI RECORD	1	82	132026	ARRESTO POSIZIONAMENTO	1	281	942445	VCE GRANO M12X20B UNI5925	1	505	561530	MOT.TL315-100-2V COMP+POMPA	1	
33	744045	RONDELLA SPECIALE D20X5	1	83	306500	DADO ES.M6 UNI5588	1	284	949420	VENTOLA MOTORE TP100 D28	1	506	691300	POMPA MEMBRANA 1TUBO 1FILETTO	1	
34	977650	VITE TCE M8X16 UNI5931	1	84	994860	VITE TE M6X40 UNI5739	1	287	790630	SPINA EL.SPIROL 8X28 DIN7343	1	507	755808	RUBINETTO ART.150/B 1/4" PORT.	1	
35	683595	PIATTAFORMA GIREVOLE RECORD	1	85	477350	GANASCIA CHIUSURA	1	289	932350	TRASFORMATORE 18W V240-415	1	508	771820	SCATOLA EL.COM.24V AUTOEST.	1	
36	130245	ANELLO DI TENUTA VRING 18	1					290	260750	CONTATTORE MC910 2450AC NO	1	509	921000	TELAIO PORTA COM.EL.24V PLAST.	1	6
37	664000	PIASTRINA CENTRALE RECORD	1	108	968920	VITE VSP M8X30 UNI5933	1	292	630700	PASSACAV.PG13.5 MIRKO RTA84213	1	510	164803	BOBINA MIN.TENS.V400	1	
38	685519	PIATTELLO GRADUATO	1	109	854100	SUPPORTO CARTER FRESA/DISCO	1	293	969950	VITE TCROCE M6X20 UNI7687	4	511	764555	SALVAMOTORE GV1 M10 4-6AMP	1	
39	527045	LEVA POSIZIONAMENTO MORSA	1	131	648720	PERNO OSCILLANTE D40	1	294	507670	GUARNIZIONE TIPO 100	1	512	787000	SPINA CEE NORM. 3P+N+T 16A	1	
40	300745	DADO ES.M8 RIBASSATO	1	133	307720	DADO ES.M8 UNI5588	1	297	306500	DADO ES.M6 UNI5588	1	513	390001	FUSIBILE VETRO 5X20	1	
41	357320	FORCELLA POSIZIONAMENTO MORSA	1	139	114745	AMMORT.GIUNTO	1	298	340516	FLANGIA ANTERIORE MOTORE	1	514	694500	PORTA FUSIBILE 1POS.	1	7
42	961045	VITE X FORCELLA DI POSIZIONAMENTO	1	140	774270	SEMIGIUNTO ELASTICO OTTONE	1									
43	918420	TARGHETTA ZERO PERFECT	1	141	534020	MANICOTTO VSF	1	301	956843	VITE TE M14X30 SINISTRA	1					
44	747920	ROSETTA PIANA D5 UNI6592	1	142	512450	GUARNIZIONE MANICOTTO TESTA	1	302	546938	MOLLA TAZZA 40X20.4X2.5	1					
45	991270	VITE TE 5X10 UNI5739	1	143	986620	VITE TE M10X60 UNI5739	1	307	335815	FLANGIA ANTERIORE MANDRINO	1					
46	176000	BUSSOLA CENTRALE RECORD	1	144	750600	ROSETTA EL.D10 DIN6798	1	309	217930	CHIAVETTA 8X7X80 UNI6604	1					
47	129470	ANELLO DI TENUTA 35X47X7	1	145	291445	CUSCINETTO 6205EE 25X52X15	1	310	129627	ANELLO TENUTA 40X52X7	1					
48	942445	VCE GRANO M12X20B UNI5925	1	146	285245	CUSCINETTO 7205B 25X52X15	1	311	288725	CUSCINETTO 6206 30X62X16	1					8
49	159190	BLOCCO MORSA SCORREVOLE RECORD	1	147	214720	CHIAVETTA 6X6X25 UNI6604	1	312	276268	CORONA M2.75 Z24	1					
50	279921	CUNEO REGISTR. MORSA L220	1	152	285245	CUSCINETTO 7205B 25X52X15	1	313	288725	CUSCINETTO 6206 30X62X16	1					

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Tavola NG 1/3

Denominazione ELENCO RICAMBI
RECORD 315/350

Gruppo
RECORD 315/350 V400

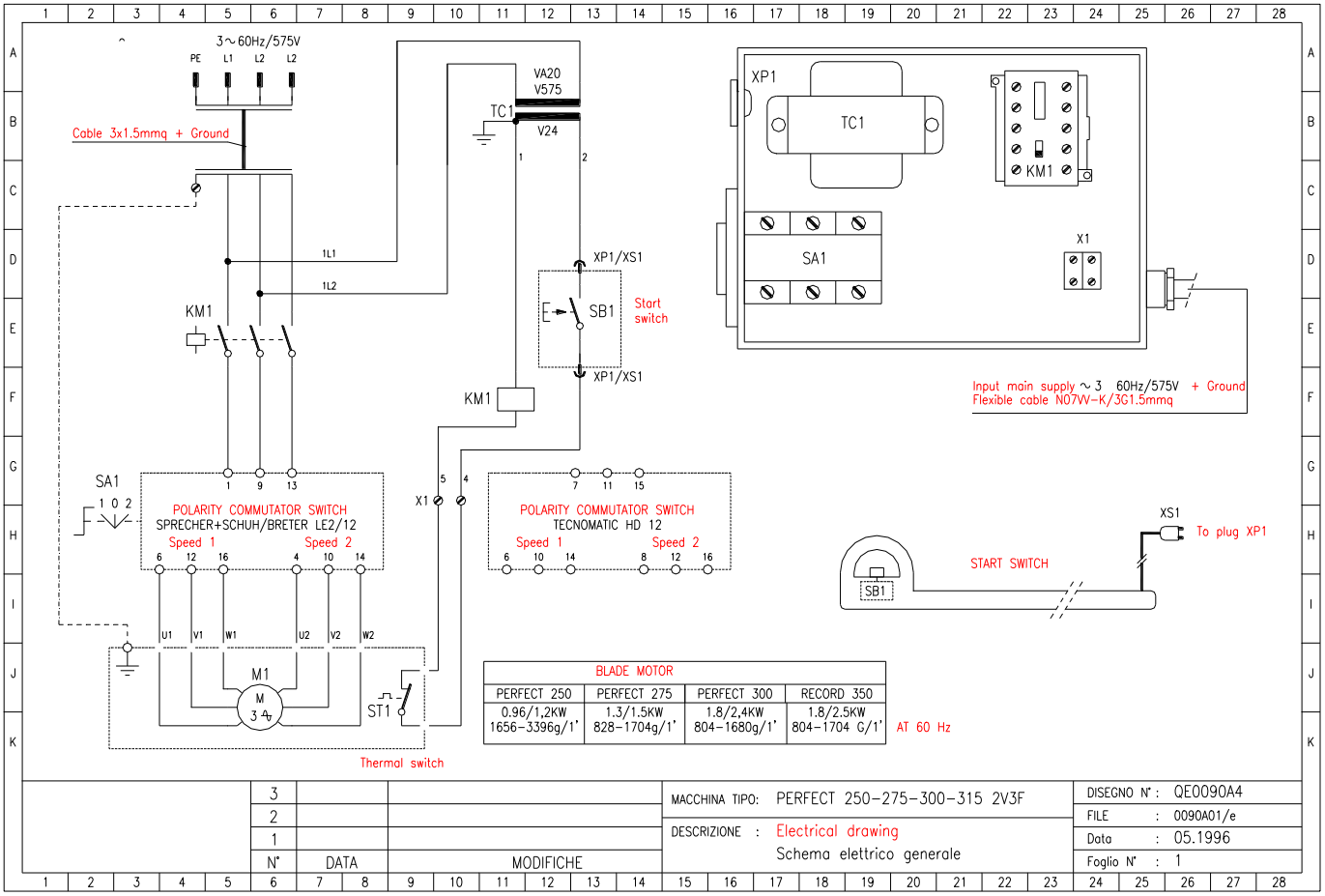
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Dis. Ng RI0143A2

09/00
Data

GANASCE H.70
Modifica



N°	DATA	MODIFICHE
3		
2		
1		

MACCHINA TIPO: PERFECT 250-275-300-315 2V3F

DESCRIZIONE : Electrical drawing
Schema elettrico generale

DISEGNO N° : QE0090A4
FILE : 0090A01/e
Data : 05.1996
Foglio N° : 1

REF.	DEVICE	SPECIFICATIONS	FUNCTION	FACTORY	TYPE	ITEM N°	Q.TY
KM1	Control master relay	4KW - 9A - 24Vac	Power circular motor	LOVATO	MC9-10-24	260750	1
				TELEMECANIQUE	LC1-K09-10		1
				SIEMENS	3TF 201E		1
SA1	Polarity commutator switch	3 Poles/16A-3 pos. Δ-0-YY	Set speed circular motor	TECNOMATIC	HD12 X222 R922	257909	1
				BRETER-SPREC.SCHUH	LE 2-12		1
M1	3 Phase motor	0.8/1KW-1500/3000 g/1'	Circular motor for PERFECT 250	ELETTROADDA	tipo 80 P70	B91712 / V575	1
		1.1/1.3KW-750/1500 g/1'	Circular motor for PERFECT 275	ELETTROADDA	tipo 90 P125	G91715 / V575	1
		1.5/2KW-750/1500 g/1'	Circular motor for PERFECT 300	ELETTROADDA	tipo 90 P135	K91717 / V575	1
		1.8/2.5KW-750/1500 g/1'	Circular motor for RECORD 315	ELETTROADDA	FC100 4/8POLES	585745 / V575	
TC1	Control circuit transformer	20VA-Vi380/575 Vo24 CSA UL	Main auxiliary supply	C.E.spa	TM VA20 575 CSA	931280	1
SB1	Pushbutton	6A 250V	Control circular motor	BREMAS	AP 65	520821	1
ST1	Thermal switch	Classe B	Control overcurrent motor	TERMIK			1
XP1	Plug	2 Poles DIN 41529	Connection start switch		PUNTO LINEA	698504	1
XS1	Receptacle	2 Poles DIN 41529	Connection start switch		PUNTO LINEA	204504	1
X1	Terminal block	Sez. 1.5mmq.	Connection thermal switch	K.E.	35038+35004	182942+182945	2

N°	DATA	MODIFICHE
3		
2		
1		

MACCHINA TIPO: PERFECT 250-275-300-315 -350 2V3F

DESCRIZIONE : General list of electrical equipments
Elenco generale materiale elettrico

DISEGNO N° : QE0090A4
FILE : 0090A02/e
Data : 05.1996
Foglio N° : 2

MAINTENANCE

DAILY
-Clean all chips from machine
-Replenish coolant level
-Inspect all blades for wear, and send full blades out for resharpening

WEEKLY
-Clean all chips from machine. Thoroughly clean vise and pivot area, and lubricate with quality lubrication
-Clean & lubricate vise area via

GEARBOX
After first 100 operating hours, drain and replace the oil in gearbox.
After every successive 1000 operating hours, again drain and replace the oil.
Recommended Gearbox Oil may be chosen from one of the following:
Oil level is correct when you can see oil halfway in the site glass with sawframe down.

RECOMMENDED LUBRICATION
Gearbox – iSO Grade 220 or SAE Grade 90
Capacity: .5 liters or 14 oz.

VISE
The vise is constructed to slide on prismatic ways which are adjustable for wear. By loosening the lock nuts along the sides and adjusting equally on each of the studs you may take up for wear. Note that you should position the vise nut adjacent to each of the adjustment points before adjusting. **Do not overtighten**, as the vise will then work with great difficulty and excessive wear will occur.

TROUBLESHOOTING

Aside from obvious machine malfunctions, most sawing problems can be attributed to improper practices or applications, and will result in blade problems.

1. Improper application, i.e. an application not suited to a circular sawing machine
2. Improper blade selection, including
   -wrong diameter
   -wrong pitch
   -wrong grind
   -dull blade
3. Saw blade not properly mounted on clean spindle
4. Material improperly clamped in machine vise
5. Improper use of coolant
6. Feed rate too heavy
7. Improper machine maintenance, especially of spindle, flanges, or vises
8. Improper training of saw operator

POSSIBLE MACHINE PROBLEMS

MOTOR FAILS TO DEVELOP FULL POWER
1. Power line overloaded
2. Undersize wire
3. Low voltage from Power Company

MOTOR OVERHEATS
1. Motor overloaded
2. High or low power supply voltage
3. Air circulation over motor restricted

COOLANT FLOW INADEQUATE
1. Valve closed or clogged
2. Coolant tank empty
3. Crimped or restricted coolant line
4. Defective valve
5. Defective or clogged pump
6. Clogged pump filter inside main tank

CANNOT MAKE SQUARE CUT

1. Improperly sharpened blade
2. Dull blade
3. Wrong blade for application;
4. Warped blade
5. Blade not properly installed on spindle
6. Chips between blade and flanges
7. Vise not set at 90 °
8. Material not clamped squarely

BLADE WILL NOT CUT AT ALL
Blade mounted with teeth in wrong direction

MOTOR WILL NOT RUN
1. Low voltage
2. Fuse blown or circuit breaker open
3. Speed selector switch set to "O"
4. Switch in handle defective
5. Open circuit in wiring
6. Motor defective

MOTOR WILL NOT START; FUSES BLOW
1. Short circuit in line or cord
2. Short circuit in motor terminal box
3. Fuses too light
4. Defective motor

MOTOR STALLS EASILY
1. Clutch Problems, including:
   -improperly cleaned and/or maintained
   -damaged, due to improper cleaning & maintenance
   SEE PAGE ON BLADE SELECTION FOR CLUTCH MAINTENANCE!
2. Low voltage supply
3. Inadequate wiring
4. Excessive pressure being used while sawing
5. Dull blade (in conjunction with above)

PREMATURE DULLING OR BREAKING OF SAWBLADE
1. Cutting speed too fast for material being cut
2. Feed rate too heavy
3. Improper blade for material being cut
4. Improper sharpening of blade for application
5. Cutting without proper coolant
6. Wrong coolant for job
7. Fusion of chips in sawblade (coolant problem)
8. Fusion of chips in sawblade (incorrect pitch)
9. Material loose in vise

SAW BLADE "KICKS"

Chips have fused in teeth (see 8 & 9 above)